



Cyber Security & Digital Forensics

CSE - 4207

Lecture - 07

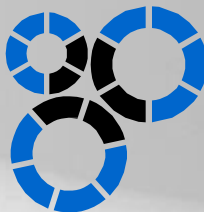
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Outline

- Transposition Ciphers
 - Rail Fence Cipher
 - Columnar Transposition Cipher
- Product Ciphers
- Rotor Machines
- Steganography



◉ Transposition Ciphers

- now consider classical **transposition or permutation ciphers**
- these hide the message by rearranging the letter order
- without altering the actual letters used
- can recognize these since have the same frequency distribution as the original text

○ Rail Fence cipher

- Encryption:
- In a transposition cipher, the order of the alphabets is rearranged to obtain the cipher-text.
- In the rail fence cipher, the plain-text is written downwards and diagonally on successive rails of an imaginary fence.
- When we reach the bottom rail, we traverse upwards moving diagonally, after reaching the top rail, the direction is changed again. Thus the alphabets of the message are written in a zig-zag manner.
- After each alphabet has been written, the individual rows are combined to obtain the cipher-text.

◉ Rail Fence Cipher Example

- For example, if the message is “DEMONSLAYER” and the number of rails = 3 then cipher is prepared as:

D			N			Y	
	E	O		S	A		E
	M			L			R

Cipher Text: DNYEOSAEMLR

◉ Rail Fence Cipher Example

- Decryption:
- As we've seen earlier, the number of columns in rail fence cipher remains equal to the length of plain-text message. And the key corresponds to the number of rails.
- Hence, rail matrix can be constructed accordingly. Once we've got the matrix we can figure-out the spots where texts should be placed (using the same way of moving diagonally up and down alternatively).
- Then, we fill the cipher-text row wise. After filling it, we traverse the matrix in zig-zag manner to obtain the original text.

◉ Rail Fence Cipher Example

Cipher Text: DNYEOSAEMLR

Cipher text length = 11, Key= 3. So Matrix=33

D			→	N				Y		
	E		O		S		A		E	
		M				L				R

Plain Text: DEMONSLAYER

◉ Columnar Transposition Ciphers

- **Encryption:**

- In a transposition cipher, the order of the alphabets is re-arranged to obtain the cipher-text.
- The message is written out in rows of a fixed length, and then read out again column by column, and the columns are chosen in some scrambled order.
- Width of the rows and the permutation of the columns are usually defined by a keyword.

Columnar Transposition Ciphers

- **Encryption:**

- For example, the word GUITAR is of length 6, and the permutation is defined by the alphabetical order of the letters in the keyword. In this case, the order would be “2 6 3 5 1 4”.
- Any spare spaces are filled with nulls or left blank or placed by a character (Example: _).
- Finally, the message is read off in columns, in the order specified by the keyword.

Columnar Transposition Ciphers

Plaint Text: Bohemian Rhapsody

Key: GUITAR (length=6)

G	U	I	T	A	R
2	6	3	5	1	4
B	o	h	e	m	i
a	n		R	h	a
p	s	o	d	y	-

- The numbering of the key is done according to the alphabetic order.
- Now print the columns in 1, 2, 3, 4, 5, 6 order
- **Cipher Text: mhyBaph oia-eRdons**

Columnar Transposition Ciphers

- **Decryption:**
- To decipher it, the recipient has to work out the column lengths by dividing the message length by the key length.
- Then, write the message out in columns again, then re-order the columns by reforming the key word.

◉ Product Ciphers

- ciphers using substitutions or transpositions are not secure because of language characteristics
- hence consider using several ciphers in succession to make harder, but:
 - two substitutions make a more complex substitution
 - two transpositions make more complex transposition
 - but a substitution followed by a transposition makes a new much harder cipher
- this is bridge from classical to modern ciphers

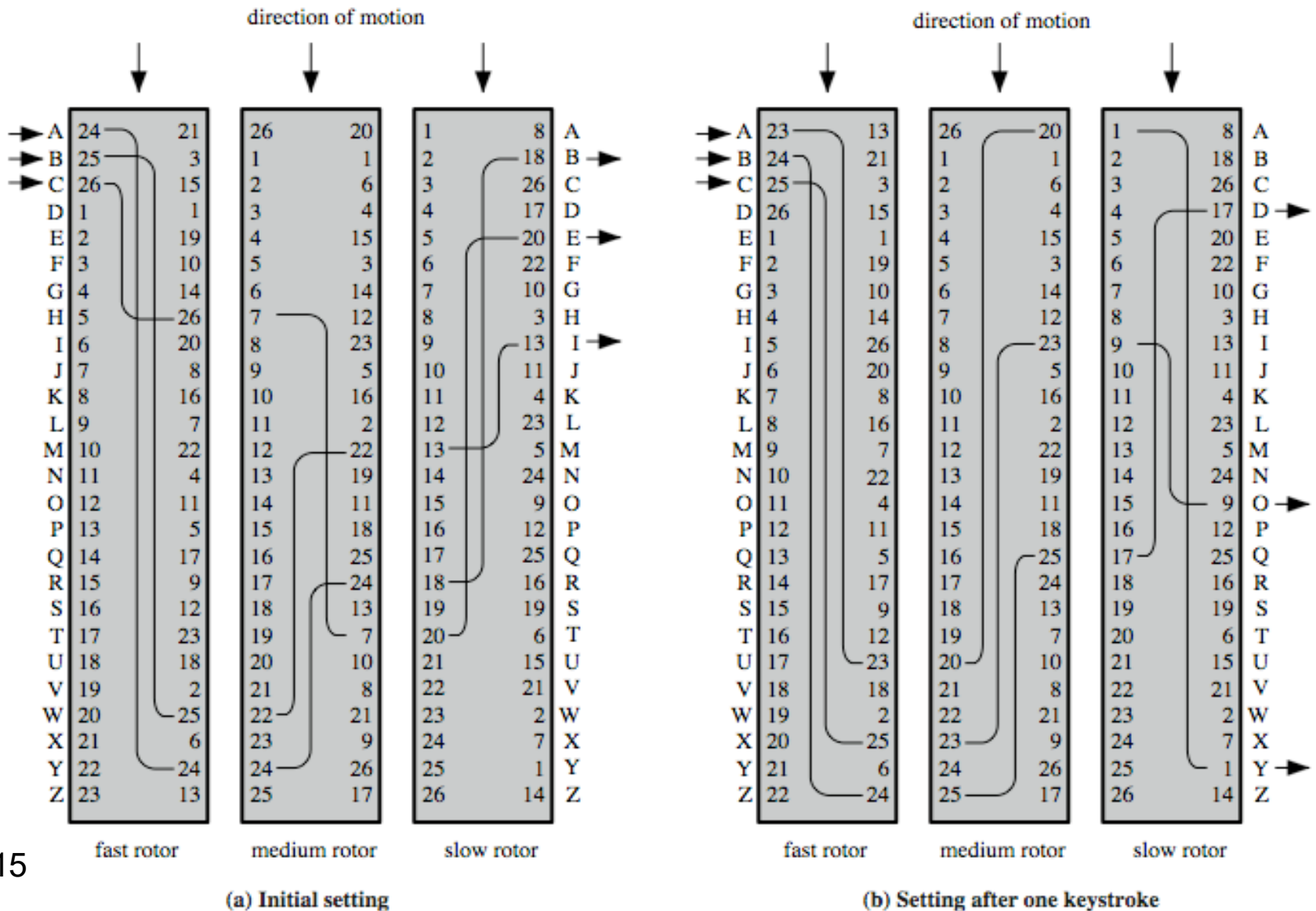
Rotor Machines

- before modern ciphers, rotor machines were most common complex ciphers in use
- widely used in WW2
- German Enigma, Allied Hagelin, Japanese Purple
- implemented a very complex, varying substitution cipher
- used a series of cylinders, each giving one substitution, which rotated and changed after each letter was encrypted
- with 3 cylinders have $26^3=17576$ alphabets

⦿ Hagelin Rotor Machine



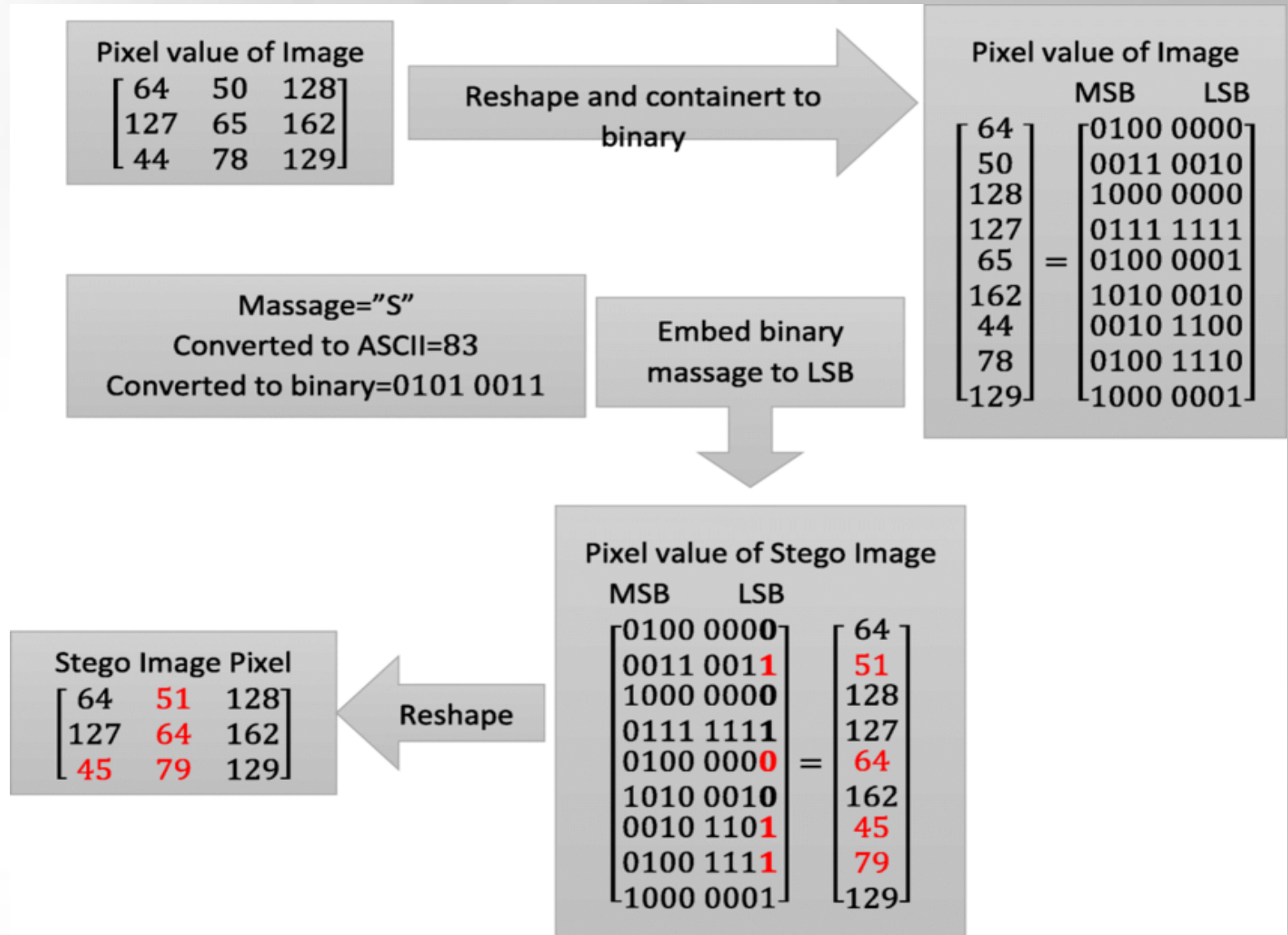
Rotor Machine Principles



◉ Steganography

- an alternative to encryption
- hides existence of message
- using only a subset of letters/words in a longer message marked in some way
- using invisible ink
- hiding in LSB (Least Significant Bit) in graphic image or sound file
- has drawbacks
- high overhead to hide relatively few info bits
- advantage is can obscure encryption use

LSB Steganography





Thank You

